

9.15 (a) From Problem 6.21, we can see that the lowest non-vanishing moments are magnetic dipole and electric quadrupole. For a single dipole, its magnetic moment is $\vec{m} = \frac{1}{2} \vec{r} \times \vec{v}$, which is

$\vec{m} = \frac{1}{2} \vec{r} \times (\vec{\omega} \times \vec{r}) = \frac{1}{2} \omega r a \hat{z} \times (\hat{z} \times \vec{r}) = -\frac{1}{2} \omega r a \vec{r} = -\frac{1}{2} \omega r a (\hat{x} \cos \phi + \hat{y} \sin \phi)$. Therefore, the magnetic dipole is

$$\begin{aligned} \vec{m} &= -\frac{1}{2} \omega r a \left[(\hat{x} \cos(\omega t) + \hat{y} \sin(\omega t)) - (\hat{x} \cos(\pi + \omega t) + \hat{y} \sin(\pi + \omega t)) \right] \\ &= -\omega r a (\hat{x} \cos(\omega t) + \hat{y} \sin(\omega t)) = \text{Re} \left[\omega r a (-1, -i, 0) e^{-i\omega t} \right] \end{aligned}$$

For the quadrupole moment, using Problem 4.2, we can write the charge density of the system as

$$\rho(\vec{x}) = -(\vec{p} \cdot \nabla) \left[\frac{1}{r^2} \delta(r-a) \delta(\cos \theta) (\delta(\phi - \omega t) - \delta(\phi - \omega t - \pi)) \right]$$

and the quadrupole moment is

$$\begin{aligned} Q_{\alpha\beta} &= \int (3x_\alpha x_\beta - r^2 \delta_{\alpha\beta}) \rho(\vec{x}) d^3x \\ &= \int \frac{1}{r^2} \delta(r-a) \delta(\cos \theta) [\delta(\phi - \omega t) - \delta(\phi - \pi - \omega t)] \left(p \frac{\partial}{\partial z} \right) (3x_\alpha x_\beta - r^2 \delta_{\alpha\beta}) d^3x \end{aligned}$$

It is straightforward to verify that only Q_{xz} and Q_{yz} are non-zero, which are

$$\begin{aligned} Q_{xz} &= \int \frac{1}{r^2} \delta(r-a) \delta(\cos \theta) [\delta(\phi - \omega t) - \delta(\phi - \pi - \omega t)] 3px d^3x \\ &= 3pa [\cos(\omega t) - \cos(\pi + \omega t)] = 6pa \cos(\omega t) = \text{Re} [6pa e^{-i\omega t}] \end{aligned}$$

$$\text{and } Q_{yz} = 6pa \sin(\omega t) = \text{Re} [6ipa e^{-i\omega t}]$$

(b) With both magnetic dipole and quadrupole moments present, the radiating magnetic field is,

as given by Eqs. (9.75) and (9.44),

$$\vec{H} = -\frac{k^2}{4\pi} \vec{n} \times \left[(\vec{n} \times \vec{m}) + \frac{i\omega}{6} \vec{Q}(\vec{n}) \right] \frac{e^{ikr}}{r}$$

$$\text{Here, } \vec{Q}(\vec{n}) = 6pa \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & i \\ 1 & i & 0 \end{pmatrix} \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} = 6pa \begin{pmatrix} \cos \theta \\ i \cos \theta \\ \sin \theta e^{i\phi} \end{pmatrix}$$

$$\text{Also, } \frac{\vec{n} \times \vec{m}}{\omega p a} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \sin\theta \cos\phi & \sin\theta \sin\phi & \cos\theta \\ -1 & -i & 0 \end{vmatrix} = \hat{x}(i\cos\theta) + \hat{y}(-\cos\theta) + \hat{z}(-i\sin\theta e^{i\phi})$$

$$\begin{aligned} \text{Thus } \vec{n} \times \vec{m} + \frac{i\omega}{b} \vec{Q}(\vec{n}) &= \left[\hat{x}(i\cos\theta) + \hat{y}(-\cos\theta) + \hat{z}(-i\sin\theta e^{i\phi}) \right. \\ &\quad \left. + \hat{x}(i\cos\theta) + \hat{y}(-\cos\theta) + \hat{z}(i\sin\theta e^{i\phi}) \right] \omega p a \\ &= 2\omega p a [\hat{x}(i\cos\theta) + \hat{y}(-\cos\theta)] \end{aligned}$$

and

$$\begin{aligned} \vec{H} &= -\frac{k^2}{4\pi} \vec{n} \times \left[\vec{n} \times \vec{m} + \frac{i\omega}{b} \vec{Q}(\vec{n}) \right] \frac{e^{ikr}}{r} = -\frac{c p a}{2\pi} k^3 \vec{n} \times [\hat{x}(i\cos\theta) + \hat{y}(-\cos\theta)] \frac{e^{ikr}}{r} \\ &= -\frac{c p a}{2\pi} k^3 \left[(\hat{x} + i\hat{y})\cos\theta - \hat{z}\sin\theta e^{i\phi} \right] \cos\theta \frac{e^{ikr}}{r} \end{aligned}$$

which agrees with Jackson up to a phase factor.

(c) From Eq. (9.21),

$$\frac{dP}{d\Omega} = \frac{Z_0}{2} r^2 |\vec{H}|^2 = \frac{Z_0}{2} \frac{c^2 p^2 a^2}{4\pi^2} k^6 (2\cos^2\theta + \sin^2\theta) \cos^2\theta = \frac{1}{8\pi^2 \epsilon_0} c k^6 p^2 a^2 (\cos^2\theta + \cos^4\theta)$$

The total time averaged power radiated is

$$\begin{aligned} P &= \int \frac{dP}{d\Omega} d\Omega = \frac{1}{8\pi^2 \epsilon_0} c k^6 p^2 a^2 \cdot 2\pi \int_{-1}^1 (x^2 + x^4) dx = \frac{1}{4\pi \epsilon_0} c k^6 p^2 a^2 \cdot \frac{16}{15} \\ &= \frac{4}{15\pi \epsilon_0} c k^6 p^2 a^2 \end{aligned}$$